

Human-AI Interaction Assignment 3

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Abstract—Multimodal learning combines information from different data sources like text, audio, and video. This report compares three multimodal fusion methods on the CMU-MOSI sentiment analysis dataset using the MultiBench framework: Early Fusion, Late Fusion, and Tensor Fusion. Each model was trained for five epochs. Late Fusion achieved the highest accuracy at 72.71%, while Early and Tensor Fusion reached 43.75% and 43.14%, respectively. Despite having the most parameters, Tensor Fusion performed the worst. The results show that choosing the right fusion strategy is important, and a more complex model does not always mean better performance.

I. INTRODUCTION

Multimodal learning improves model performance by combining data from multiple sources, such as text, audio, and video. Human sentiment, for example, is expressed through spoken words, vocal tone, and facial expressions. The CMU-MOSI dataset provides aligned text, audio, and visual features from opinion videos, making it a good benchmark for multimodal sentiment analysis.

The goal of this project is to compare three fusion strategies in the MultiBench framework and evaluate how well they work for sentiment classification.

II. MATERIALS AND METHODS

A. Dataset

I used the CMU-MOSI dataset, which contains video segments with sentiment labels across three modalities:

- Text
- Audio
- Visual

I used the default MultiBench preprocessing pipeline through a custom experiment script named `assignment_mosi.py`. The script automatically loads the CMU-MOSI dataset, prepares the train/validation/test splits, trains each fusion model, and records the resulting performance metrics. The data is split into training, validation, and testing sets as shown in Table I.

Table I: CMU-MOSI Dataset Split

Subset	Samples
Training	1283
Validation	214
Testing	686

B. Fusion Methods

I tested three fusion methods available in MultiBench:

Early Fusion: This method concatenates the raw text, audio, and visual features before any feature extraction. A single GRU processes the combined features, followed by an MLP for prediction. It is simple and has fewer parameters, but it might lose modality-specific details early on.

Late Fusion: Here, each modality has its own GRU encoder. The encoded outputs are then concatenated and fed into a prediction network. This keeps the modality-specific information intact until the final fusion step.

Tensor Fusion: This method uses tensor outer products to explicitly model interactions between the modalities. It captures complex cross-modal relationships but requires significantly more parameters and computation.

C. Experimental Setup

All experiments were conducted using a custom Python script (`assignment_mosi.py`) developed for this assignment. The script uses the official MultiBench implementations of Early Fusion, Late Fusion, and Tensor Fusion, automatically trains each model for five epochs, evaluates them on the test set, and saves the results for comparison. I used the AdamW optimizer with an L1 loss function. The main evaluation metric is binary sentiment classification accuracy. I also tracked the total number of trainable parameters and the training time to compare model complexity.

III. RESULTS

A. Performance Comparison

Table II shows the results for the three methods.

Table II: Comparison of Fusion Methods

Method	Accuracy (%)	Parameters	Time (s)
Early Fusion	43.75	1,680,897	11.63
Late Fusion	72.71	2,570,311	13.42
Tensor Fusion	43.14	4,193,229	12.81

Late Fusion got the highest accuracy at 72.71%, which is much better than the other two. Early Fusion and Tensor Fusion performed similarly, with accuracies around 43%. Even though Tensor Fusion has the most parameters, it actually performed the worst.

B. Model Complexity Analysis

As expected, Early Fusion had the fewest parameters since it processes the combined features directly. Late Fusion added more parameters because of the separate encoders for each modality. Tensor Fusion was the largest model by far due to the outer product calculations. However, this extra complexity did not translate into better accuracy.

IV. CONCLUSION

This project compared three multimodal fusion methods on the CMU-MOSI dataset using the MultiBench framework through a custom implementation in `assignment_mosi.py`. Late Fusion performed the best with an accuracy of 72.71%, while Early and Tensor Fusion scored around 43%. The results show that simply adding more parameters or using a more complex fusion method like Tensor Fusion doesn't guarantee better performance. For this dataset, Late Fusion provided the best balance of accuracy and model design.